German University in Cairo
CSEN 903 · Advanced Computer Lab · Winter 2026

Formula 1 Podium Prediction

Milestone 1 · Analytical Report

Team 108

Mohamed Karim · 58-1483
Aly Talaat · 58-4123
Ahmed Nader · 58-1824
Youssef Abdelazim · 58-0684

October 2026

Contents

1. Task and data 3

2. Data cleaning and auditing 3

2.1 EDA before cleaning . . . 3

2.2 Sentinels and semantic typing . . . 5

2.3 Key integrity . . . 5

2.4 Grain fix: one row per (raceId, driverId) . . . 5

2.5 Scope decisions . . . 6

2.6 EDA after cleaning . . . 6

3. Data-engineering questions 7

Q1. Front-row conversion by circuit, and the change after 2014 . . . 7

Q2. Home advantage after controlling for starting position . . . 8

Q3. Mechanical-retirement rate by constructor, 2014–2021 vs 2022–2024 . . . 9

4. Feature selection and engineering 11

4.1 Leakage control at the pre-race cutoff . . . 11

4.2 Train / validation / test split . . . 11

4.3 Evidence for every candidate (training seasons only) . . . 12

4.4 Selected features and why each is valid at the pre-race cutoff . . . 13

5. Pre-processing 14

5.1 Missing values: remove or impute? . . . 14

5.2 Preprocessing-order experiments . . . 15

6. Predictive modeling 15

6.1 Metrics and threshold . . . 15

6.2 Baseline and modeling attempts . . . 15

6.3 Final comparison on train, validation and the unseen test seasons . . . 17

7. Experiment: feature-group ablation 18

8. Explainability (XAI) 19

8.1 Global: permutation importance . . . 19

8.2 Global: SHAP . . . 19

8.3 Local: one driver at a time (force, waterfall, LIME) . . . 20

8.4 Explanations are not causal effects . . . 22

9. Inference function 22

10. Known limitations 23

1. Task and data

- **Unit of analysis:** one driver entering one race. **Target:** `podium` = 1 if `positionOrder` ≤ 3.
- **Prediction cutoff:** after qualifying, before the race starts. Anything decided in the race itself (results, status, points, laps, pit stops, post-race standings) is **future information** and is never used as a feature.
- **Data:** the Kaggle/Ergast dataset, 14 CSV tables (results, races, drivers, constructors, circuits, qualifying, standings, status, …). The raw files are read as text and **never modified**; all cleaning happens on copies.
- **Notebook:** one run-all Kaggle notebook, Parts 1–7 (cleaning → questions → features → models → ablation → XAI → inference). All figures in this report come from that run.

2. Data cleaning and auditing

2.1 EDA before cleaning

We first measured the main data problems with numbers and plots rather than by eye.

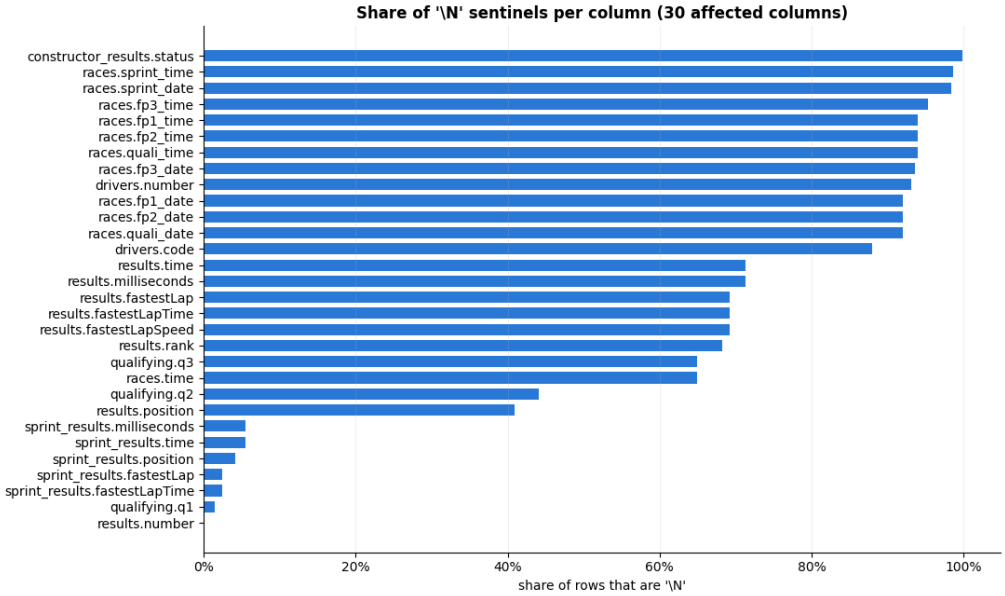


Figure 1. Where the \N sentinels are (160,076 cells in total). They sit in post-race fields (fastest lap, milliseconds, position), in qualifying q2/q3 (knocked-out drivers) and in session dates that only exist for recent seasons.

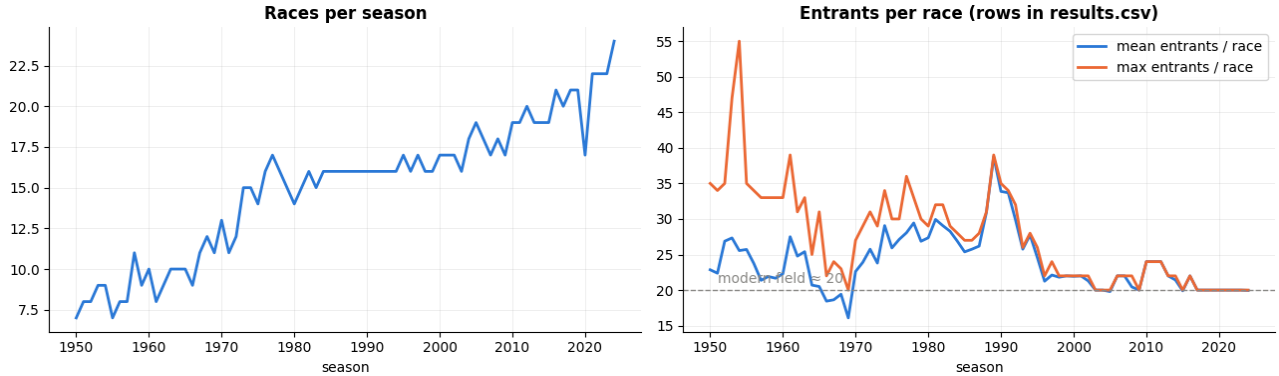


Figure 2. The sport changed shape: the calendar grew from 7 to 24 races, and rows per race peaked at 35–40 around 1989 because non-qualifiers still have a row. Those rows can never be a podium, so they dilute the class balance.

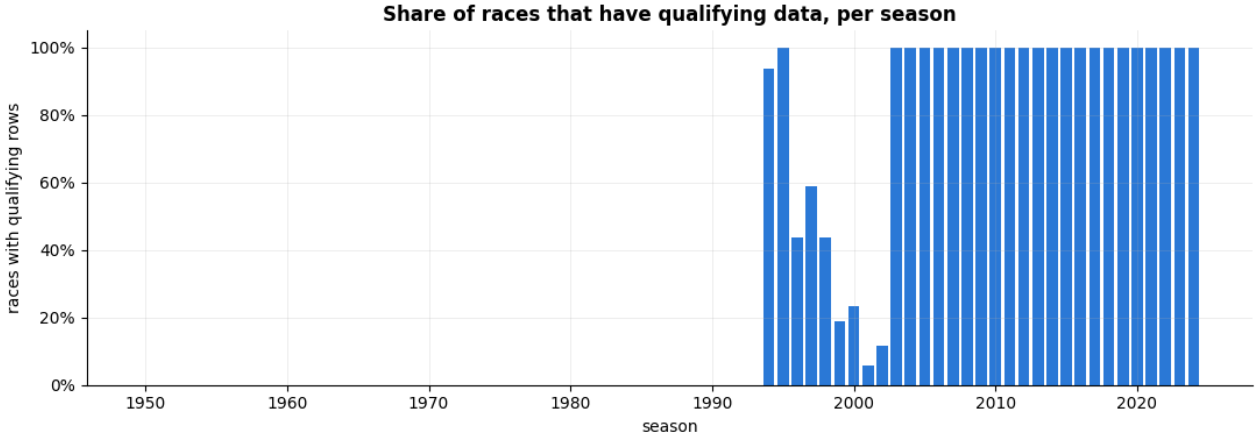


Figure 3. Qualifying-coverage drift: `qualifying.csv` has nothing before 1994, only 6–59% of races in 1996–2002, and every race from 2003. This later decides how the qualifying feature is handled (a "no qualifying time" flag, Section 5).

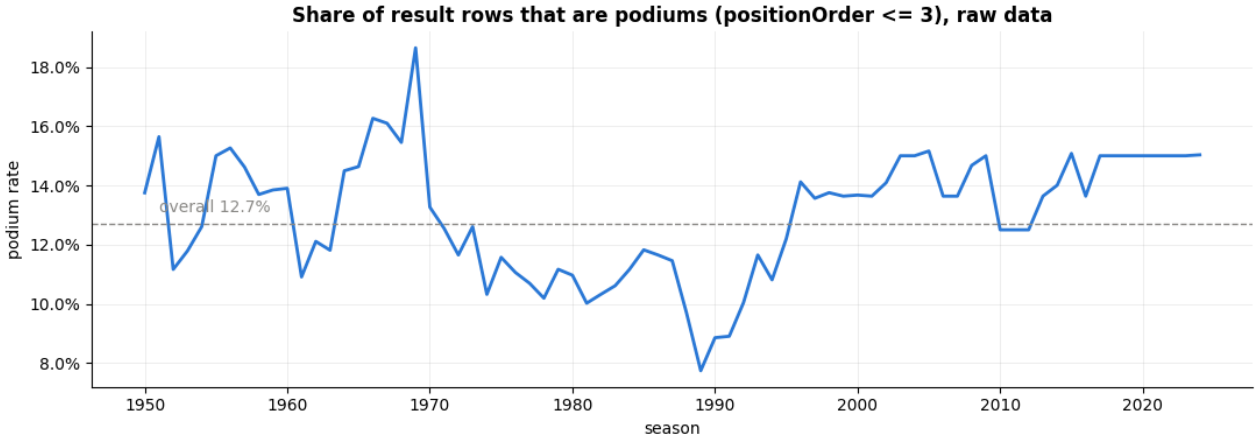


Figure 4. Podium rate per season in the raw data. Only ≈12.7% of rows are podiums (imbalanced), so accuracy would be misleading; we use PR-AUC and F1. The dip in the late 1980s comes from non-qualifiers that still have a row.

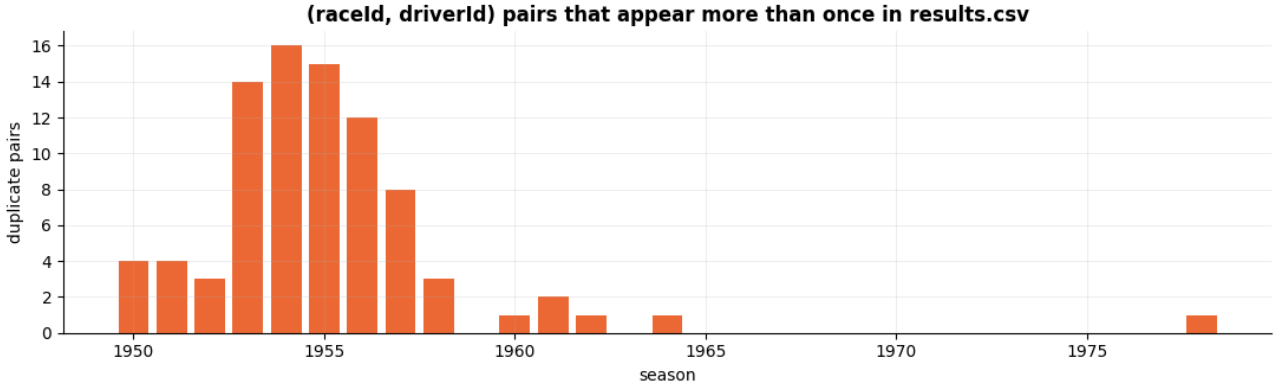


Figure 5. Duplicate driver-race rows: 176 rows form 85 duplicated (`raceId`, `driverId`) pairs, almost all in the 1950–1964 shared-drive era (latest 1978).

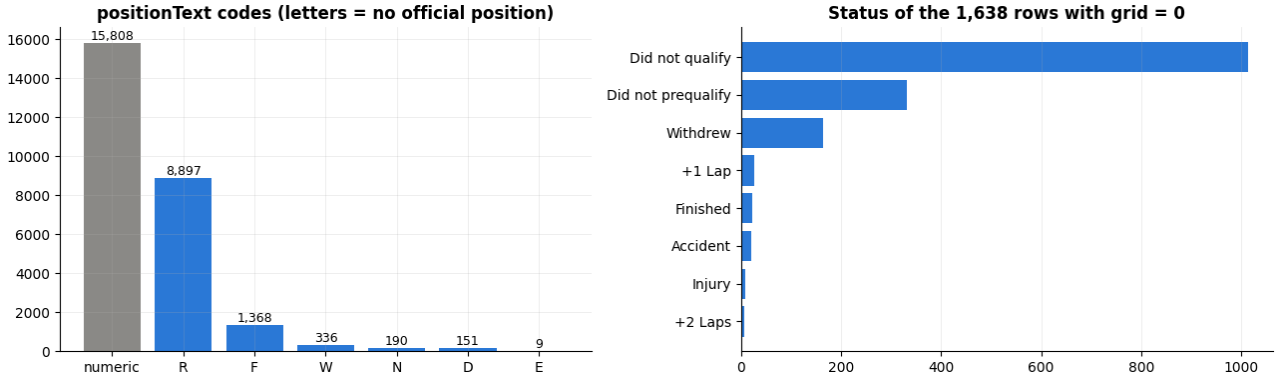


Figure 6. Rows that never started the race: F (failed to qualify, ≈1.4k rows) and W (withdrawn) never took the start. `grid = 0` is a mix: mostly non-starters, but some finished the race (genuine pit-lane starts), so non-starters are removed by status, not by `grid` (Section 2.5).

2.2 Sentinels and semantic typing

- **Sentinels:** `\N` → missing (NaN) in every table; a column becomes numeric **only if every non-missing value is a number**, so codes such as `positionText = "R"` stay text. Sentinel cells: 160,076 → 0.
- **Text consistency:** one trailing space (`"Argentinian "`) and one country spelled two ways (`"United States"` vs `"USA"`) were fixed, because Q2 matches nationality to circuit country.
- **Semantic types:** dates → `datetime`; lap, qualifying and pit-stop durations (`"1:26.572"`) → seconds. The parser is verified against the millisecond columns of `lap_times` and `pit_stops` (maximum difference 0.0 s), so it can be trusted on qualifying times. `results.time` is not parsed (it mixes total times and gaps).

2.3 Key integrity

All **14 primary keys** are unique and non-null, and **all 23 foreign keys reconcile** (0 orphan values), so joins cannot silently lose rows. The only failure is the natural key of `results` (`raceId`, `driverId`): the 91 extra rows of the shared drives.

2.4 Grain fix: one row per (raceId, driverId)

A shared car is stored as one row per driver with the car's grid slot copied, so a driver who took over a teammate's car mid-race carries the teammate's grid slot. We therefore make **two separate decisions**:

Part of the row	Taken from	Why
Features (grid, constructor, …)	the car the driver **started in**: the row on a grid slot he alone owns; if inconclusive, the lowest `resultId` (entry order)	Only the starting car is known at the prediction moment; a car taken over mid-race is future information (leakage).
Label (`podium`)	the driver's **credited** result in any of his cars that race	The label is the answer, so post-race information is allowed; e.g. Moss's credited 1957 win is kept.

Table 1. The grain rule. Rows 26,759 → 26,668; (`raceId`, `driverId`) is now unique and no driver-race was lost.

2.5 Scope decisions

- **Indianapolis 500 (1950–60) excluded:** only 0.5% of its drivers raced any other Grand Prix that season and its 33-car oval field is a different population; it would distort the home-race analysis (Q2). 11 races of 1,125.
- **Never qualified → removed** (did not qualify/pre-qualify, 107% rule, withdrawn without a grid slot): they are not on the grid, so we would never make this prediction. **Qualified but did not start → kept** with `did_not_start` = 1 (e.g. Leclerc's pole at Monaco 2021): at prediction time they were on the grid, and removing them would use post-cutoff information. Pit-lane starters are kept.

2.6 EDA after cleaning

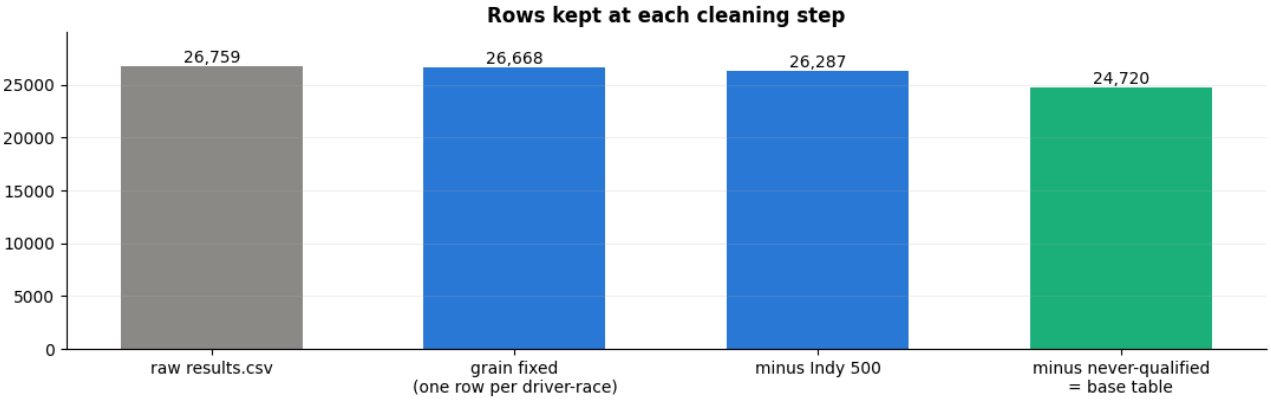


Figure 7. Row funnel from the raw results to the clean base table: 2,039 rows removed (7.6%), 24,720 driver-races left.

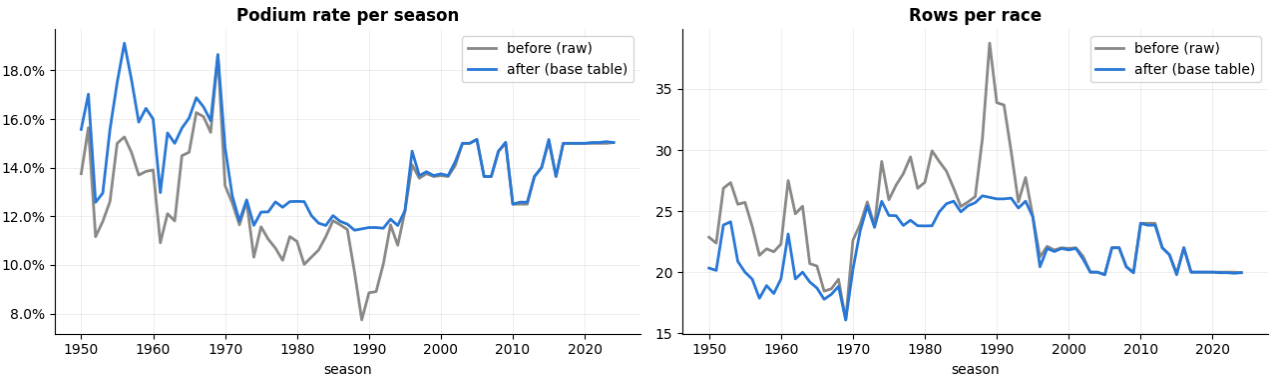


Figure 8. Before vs after cleaning. Removing non-qualifiers brings the late-1980s fields back to ≈26 real starters, and the podium rate becomes much more stable across eras (overall 12.7% → 13.6%).

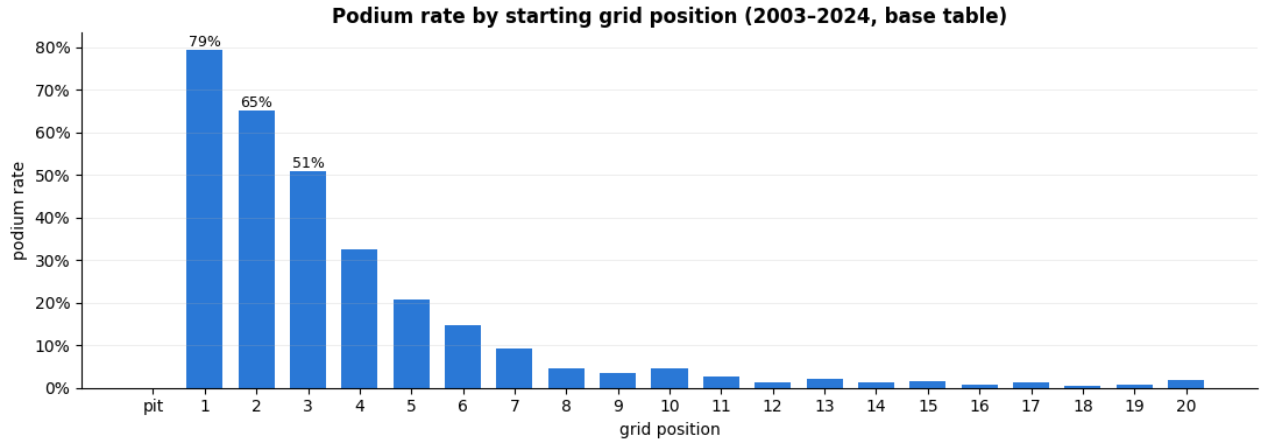


Figure 9. The strongest single signal (2003–2024): pole converts to a podium ≈80% of the time, P3 ≈50%, P5 ≈20%, P10 ≈5%. The steep, non-linear shape matters for model choice.

Missing standings rows: 422 driver-races have no standings row; **100%** of them belong to drivers with 0 season points (up to 2011 the table omits drivers who have not scored). A missing prior-standings value therefore means **0 points**, not "unknown".

3. Data-engineering questions

All three answers start from the clean driver-race table and add other tables through multi-hop joins (every join is validated so it cannot duplicate rows). Every rate is shown next to its count, and groups that are too small are not ranked.

Q1. Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Join path: results → qualifying → races → circuits. **Front row** = P1 or P2, measured both by **qualifying position** and by **grid** (after penalties and, in 2021–22, sprints). **Seasons 2003–2024**, where qualifying coverage is complete, so both versions use the same races; eras 2003–2013 vs 2014–2024 (11 seasons each). A circuit needs ≥ 16 front-row starts to be ranked overall and ≥ 8 per era.

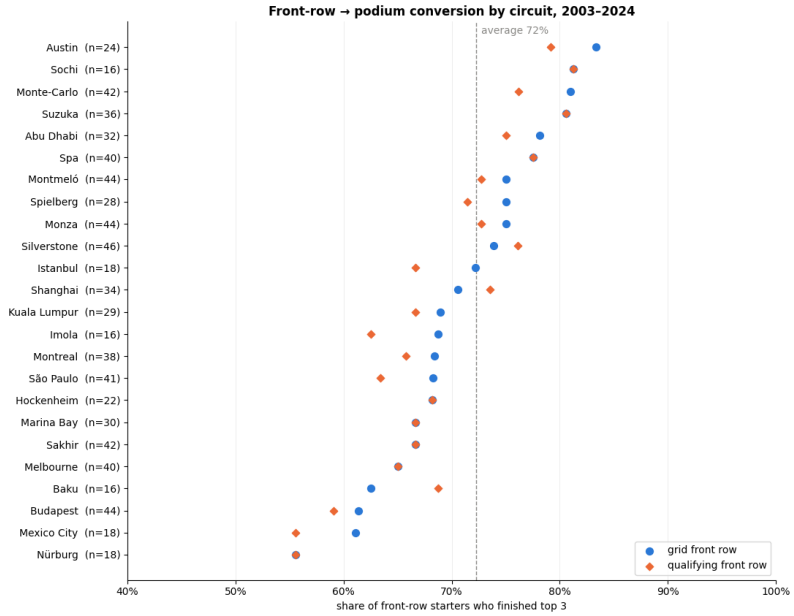


Figure 10. Front-row conversion per circuit, 2003–2024 (blue = grid definition, orange = qualifying definition, dashed = average). n = front-row starts behind each rate.

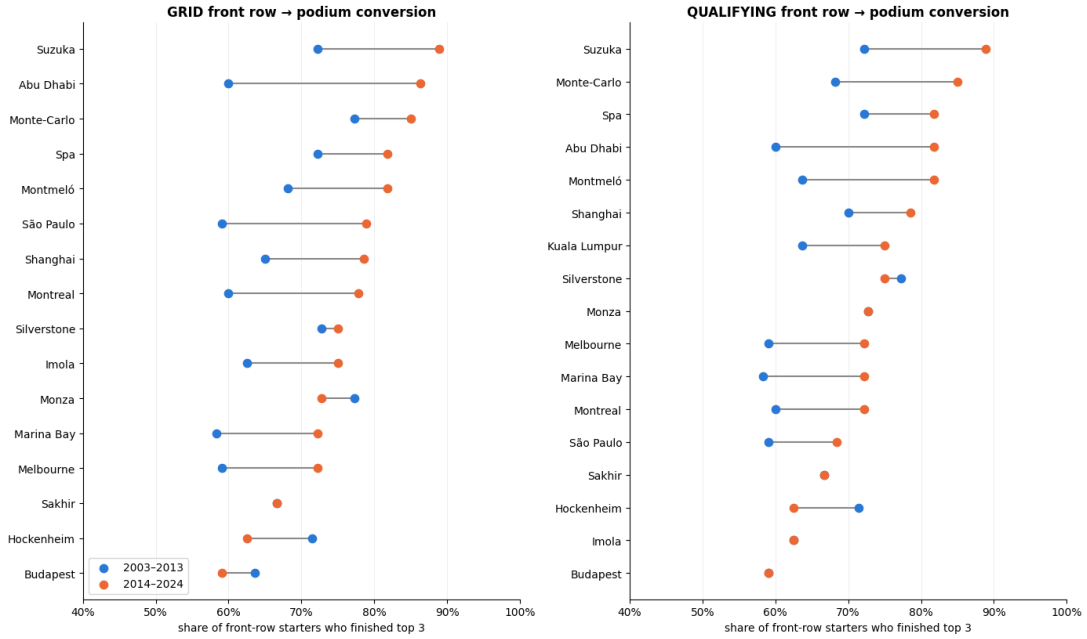


Figure 11. Ranking change between eras: each line is one circuit (blue = 2003–2013, orange = 2014–2024).

Answer

- **Overall:** 72.2% of front-row starters finish on the podium (70.8% by qualifying position). Highest: **Austin 83%** (20 of 24), Sochi 81%, **Monaco 81%** (34 of 42), Suzuka 81%. Lowest: Nürburgring 56% (10 of 18), Mexico City 61%, Budapest 61%.
- **Yes, the ranking changes a lot after 2014.** Top 3 in 2003–2013: Monaco and Monza (77%, tied), Silverstone (73%). In 2014–2024: **Suzuka (89%)**, Abu Dhabi (86%), Monaco (85%). A circuit moved **4.8 places** on average (of 16); only 3 of the top 5 stayed. Biggest moves: Abu Dhabi 12th → 2nd, Monza 1st → 11th. Conversion rose at 12 of 16 circuits. **Monaco** is the only circuit in the top 3 of both eras, consistent with being the hardest track to overtake on.
- **Qualifying vs grid:** 37 of 855 front-row qualifiers (4.3%) did not start from the front row (penalties, sprints). Conversion is slightly lower by the qualifying definition, because penalised qualifiers podium less, so `grid` **is the better model feature**.
- **Grain / missingness / cutoff:** one row per driver-race; 22 rows (0.25%) lack a qualifying position and are dropped from the qualifying version only; grid and qualifying are both known before the race. With 8–24 front-row starts per circuit and era, part of the reshuffle is small-sample noise.

Q2. Do drivers podium more often at home, after controlling for starting position?

Join path: results → drivers (nationality) → races → circuits (country). The two tables use different words ("British" vs "UK"), so we built an explicit, documented **nationality → country map** (every one of the 43 labels has a decision; 3 dual-nationality rows excluded). **Control:** home and away drivers are compared **only within the same grid group** (P1, P2, P3, P4–6, P7–10, P11+/pit).

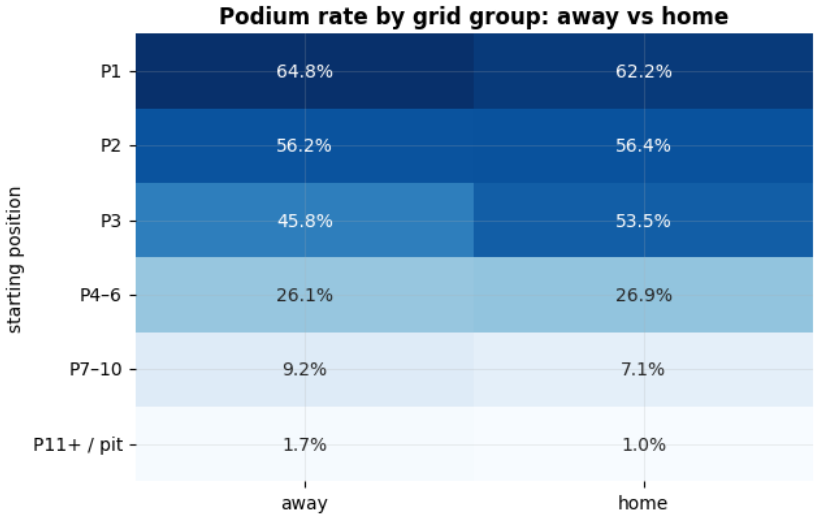


Figure 12. Podium rate by grid group, away vs home (pivot table shown as a heatmap). If racing at home helped, the home column would be higher in every row.

Grid group	Home starts	Home rate	Away starts	Away rate	Difference
P1	82	62.2%	1,034	64.8%	−2.6 pts
P2	78	56.4%	1,034	56.2%	+0.2 pts
P3	71	53.5%	1,045	45.8%	+7.7 pts
P4–6	227	26.9%	3,115	26.1%	+0.8 pts
P7–10	336	7.1%	4,117	9.2%	−2.1 pts
P11+ / pit	1,150	1.0%	12,428	1.7%	−0.6 pts

Table 2. Home vs away podium rate inside each grid group (1950–2024).

Answer

- **No.** After controlling for starting position, drivers do **not** podium more often at home.
- Raw comparison (no control): home drivers podium *less* (11.8% vs 13.7%), mainly because 59.2% of home starts are from P11 or further back (54.6% away).
- Inside the grid groups home is higher in 3 groups and lower in 3; the largest gap (P3, +7.7 points) is about 5 podiums out of 71 home starts. Combined: at the away rate of their own grid groups, home drivers would have scored **238.8** podiums; they scored **230** (−0.46 points per 100 starts). Starting position explains most of the raw gap.
- This is an association, not a cause: nationality is only a proxy for "home". **Cutoff:** nationality and circuit country are known before the race, so `home_race` is a legitimate candidate feature (it is later dropped for having no signal, Section 4).

Q3. Which constructors have the highest mechanical-retirement rate, 2014–2021 vs 2022–2024?

Join path: results → status → races → constructors. All **139 status values** are mapped to one category (mechanical 68, finished 34, ambiguous 16, accident/driver 11, regulatory/other 10; an assert stops the notebook if one is unmapped). Ambiguous labels (puncture, "Retired", loose wheel) are **not** counted as mechanical. **Rate = mechanical retirements ÷ race starts** (cars that never took the start excluded); teams are kept as recorded (renamed teams are separate) and need ≥ 30 starts per era.

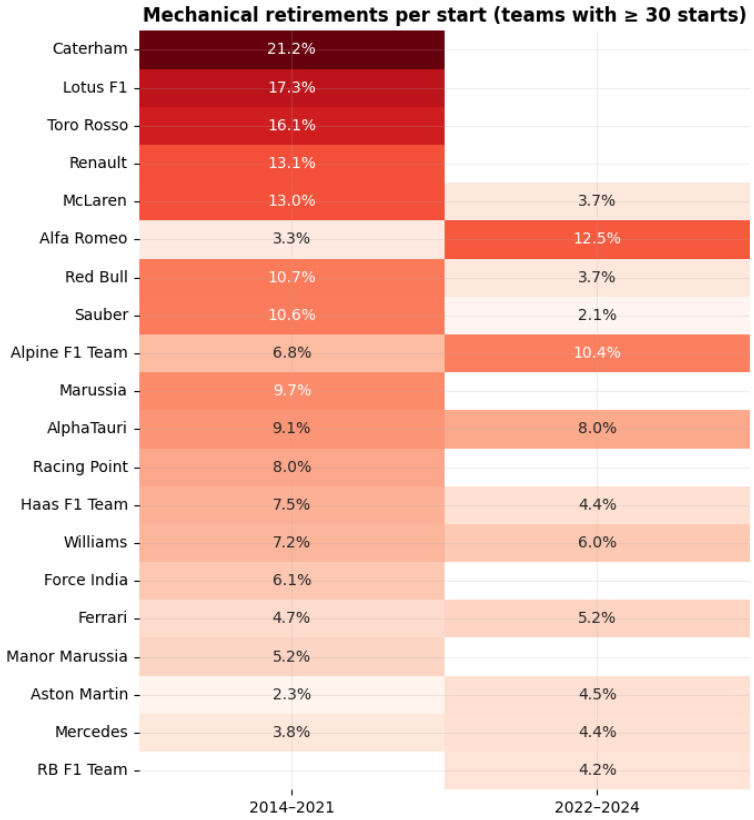


Figure 13. Mechanical-retirement rate per constructor and era (darker = more breakdowns per start; blank = team did not exist under that name or had < 30 starts).

Answer

2014–2021: Caterham (21.2%), Lotus F1 (17.3%) and Toro Rosso (16.1%). 2022–2024: Alfa Romeo (12.5%), Alpine (10.4%) and AlphaTauri (8.0%).

Rank	2014–2021	2022–2024
1	Caterham **21.2%** (7 of 33; one season only)	Alfa Romeo **12.5%** (11 of 88)
2	Lotus F1 **17.3%** (13 of 75)	Alpine **10.4%** (14 of 135)
3	Toro Rosso **16.1%** (39 of 242)	AlphaTauri **8.0%** (7 of 88)

Table 3. Highest mechanical-retirement rates (teams with ≥ 30 starts).

- Caterham's rate rests on only 33 starts (each breakdown moves it by 3 points), so it is the least certain number in the table. Among teams that raced the whole 2014–2021 period, the least reliable were Toro Rosso (16.1%), Renault (13.1%) and McLaren (13.0%).
- Reliability improved between the eras (Figure 13): the highest rate fell from 21.2% to 12.5%, and in 2022–2024 the front-running teams were the most reliable (Red Bull and McLaren 3.7%).
- **Grain / missingness / cutoff:** one row per race start (cars that never started excluded); every row has a status and all 139 status labels are classified. Retirement status is post-race: it is used only to *describe* reliability, and as a model feature it appears only as a **past** value (a team's mechanical-retirement rate over its previous 10 races).

4. Feature selection and engineering

4.1 Leakage control at the pre-race cutoff

For race *R* a feature may only use what is fixed before the lights go out (the grid and qualifying of *R*, static facts) and results of races **before** *R*. The code enforces this: races are put in time order; every form feature uses `shift(1)`; standings come from the **previous round** (the table for *R* already contains *R*'s points); team features are summed per race and then shifted, so a teammate's result in the same race cannot leak in.

Left out	Reason
`positionOrder`, `position`, `points`, `laps`, `statusId`, `time`, current fastest lap	the race result itself: leakage
`did_not_start`	decided after the grid is fixed; the row stays, never as an input
current-race `pit_stops`, `lap_times`, standings	happen during / after the race: leakage
`sprint_results`	only from 2021 (almost only validation/test); its effect is already in `grid_pos`
`year`; driver, team and circuit IDs as categories	unseen values in the test era; the model would memorise names, and teams get renamed. Their information enters through form and standings
raw qualifying times in seconds	not comparable between circuits; only the gap to the fastest is

Table 4. Fields left out as future information or as non-transferable.

Leakage test (change-a-race): we deliberately change one race's results (flip every podium, reverse the order, add 1,000 standings points, mark every car as a mechanical failure), rebuild **all** features and check: features of that race and every earlier race are **identical**, while hundreds of later rows change (448, 450 and 300 for 2010, 2016 and 2023). A test for forbidden column names also passes. Every feature therefore uses only the past.

4.2 Train / validation / test split

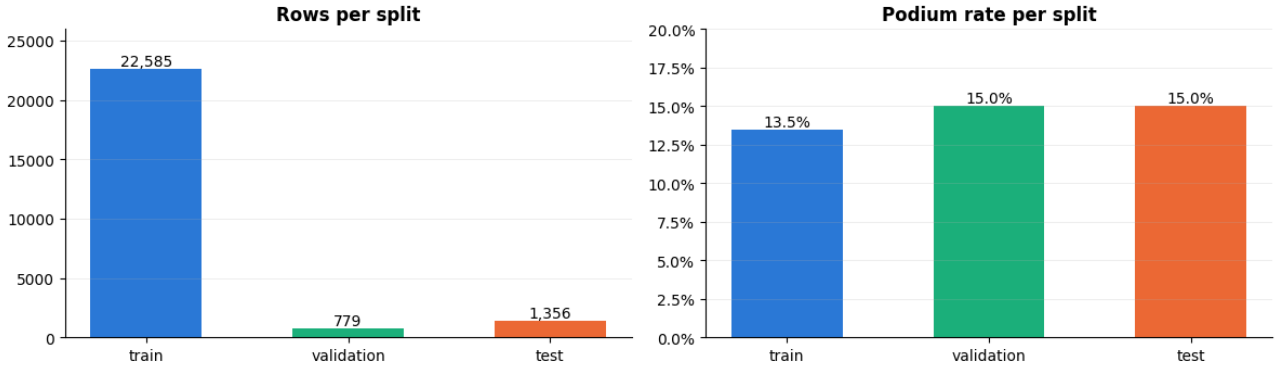


Figure 14. Split by season: train 1950–2019 (1,007 races, 22,585 rows, 13.5% podiums), validation 2020–2021 (779 rows, 15.0%), test 2022–2024 (1,356 rows, 15.0%).

- **By season, not random rows:** in real use later races are predicted from earlier ones; and the three podiums of a race depend on each other, so splitting one race's drivers between train and test would let the model partly guess test labels.
- **All seasons from 1950 in training:** 3.3× more examples than training from 2003, covering many rule eras. The cost (no qualifying times before 1994, no team standings before 1958) is handled with missing-value flags, not by deleting seasons.
- **Test = 2022–2024, a complete new rule era** (ground-effect cars), used once at the end: an honest check of whether patterns learned on older eras still work after a regulation change. The slightly lower training podium rate reflects older, bigger fields (3 ÷ 26 cars = 11.5%).

4.3 Evidence for every candidate (training seasons only)

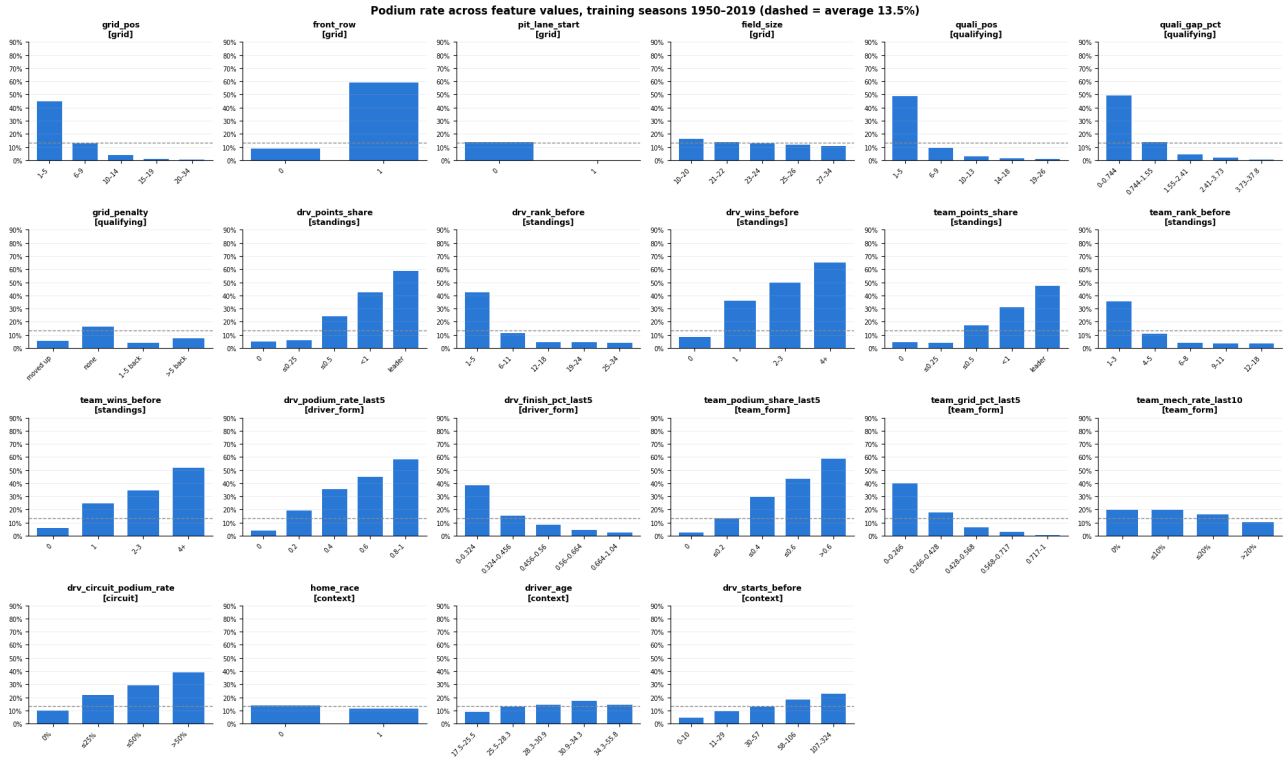


Figure 15. Podium rate across the values of each candidate (training seasons; dashed line = average). Steep: grid, qualifying, standings, driver and team form. Medium: circuit history, experience. Flat: field size, home race, driver age.

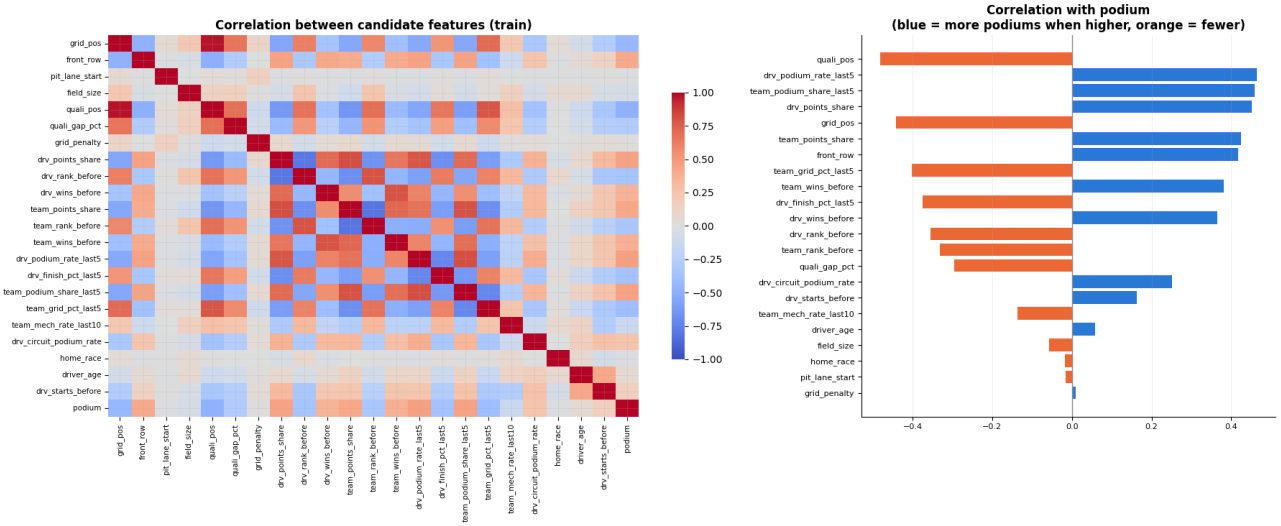


Figure 16. Correlation between candidates (left) and of each candidate with podium (right). Only one pair is a near-duplicate: `grid_pos` and `quali_pos` (0.98).

Selection rules (all on the training seasons): (1) leakage-safe; (2) |correlation with podium| ≥ 0.10, backed by a visible change of the podium rate in Figure 15; (3) not a duplicate or a copy of another feature (correlation ≥ 0.95, or computed from it). **15 of 22 candidates are kept.** Dropped: `quali_pos` (duplicate of the grid; the grid already contains penalties, Q1) and `front_row` (= grid ≤ 2); `pit_lane_start`, `field_size`, `grid_penalty`, `home_race` (−0.02, matching Q2) and `driver_age` (0.06) for too little signal.

4.4 Selected features and why each is valid at the pre-race cutoff

Group	Feature	Definition	Why it is known before the race	Corr.
grid	`grid_pos`	starting slot after penalties; pit lane = last	the grid is fixed before the start	−0.44
qualifying	`quali_gap_pct`	best qualifying lap, % slower than the fastest	qualifying ends before the race	−0.30
standings	`drv_points_share`	season points ÷ leader's points	standings after the **previous** round	+0.45
	`drv_rank_before`	championship position (no points = last)	previous round	−0.36
	`drv_wins_before`	season wins so far	previous round	+0.37
	`team_points_share`	team points ÷ leading team's points	previous round (missing before 1958)	+0.42
	`team_rank_before`	team championship position	previous round	−0.33
	`team_wins_before`	team season wins so far	previous round	+0.38
driver form	`drv_podium_rate_last5`	podium share in the previous 5 races	earlier races only (`shift(1)`)	+0.46
	`drv_finish_pct_last5`	average finishing percentile, previous 5 races	earlier races only	−0.37
team form	`team_podium_share_last5`	team podiums per car, previous 5 races	team-level shift	+0.46
	`team_grid_pct_last5`	team's average grid percentile (car pace)	team-level shift	−0.40
	`team_mech_rate_last10`	mechanical retirements per start, previous 10 races (Q3)	team-level shift	−0.14
circuit	`drv_circuit_podium_rate`	podium rate in earlier visits to this circuit	earlier races only	+0.25
context	`drv_starts_before`	number of earlier races (experience)	earlier races only	+0.16

Table 5. The 15 selected features (correlation with podium on the training seasons).

Shares and percentiles instead of raw numbers: a win was worth 8 points in the 1950s and 25 from 2010, and 12th of 24 cars is not 12th of 20; dividing by the leader or by the field size puts every season on the same 0–1 scale. **Windows:** 5 races (about a quarter season) for form; 10 for rarer mechanical failures. **Timing errors:** a qualifying lap more than twice as slow as pole (one 1995 time of 1,002 s) is treated as no valid time; left in, this one row would hide the whole qualifying signal (−0.08 instead of −0.30).

5. Pre-processing

5.1 Missing values: remove or impute?

Feature(s)	Missing means	Train rows	Podium rate when missing	Handling
`quali_gap_pct`	no qualifying time (before 1994, or a timing error)	63.7%	13.1%	median + flag `no_quali_time`
`drv_circuit_podium_rate`	first visit to this circuit	31.4%	6.7%	median + flag `first_circuit_visit`
team standings (3)	season before 1958 (no constructors' title)	5.2%	15.7%	median + flag `no_team_standings`
driver form (2)	driver's debut	3.0%	2.5%	median (debut is already `drv_starts_before` = 0)
team form (3)	team's first race (new / renamed)	1.1%	5.0%	median (too rare for a flag)

Table 6. Every gap has a known meaning (training podium rate 13.5%). In validation and test, qualifying is 99% complete.

- **No column or row is removed:** the largest gap (qualifying, 64%) has a known cause, is almost complete in the seasons we predict, and is one of the strongest signals; removing rows would delete whole eras and predictions we must be able to make (a rookie's first race).
- **Training median** (robust to extreme values) fills the gaps; a **0/1 flag** is added where the gap itself is information, so the model can tell a filled value from a real one.
- **Scaling:** `StandardScaler` gives every input mean 0 and standard deviation 1 (grid 1–34 and experience 0–324 starts would otherwise dominate the 0–1 rates in logistic regression and the neural network).
- **Every learned value (medians, means, standard deviations) comes from the training seasons only** and is applied unchanged to validation, test and new races. Order: flags → fill → scale. Result: 18 model inputs (15 features + 3 flags), 0 missing values.

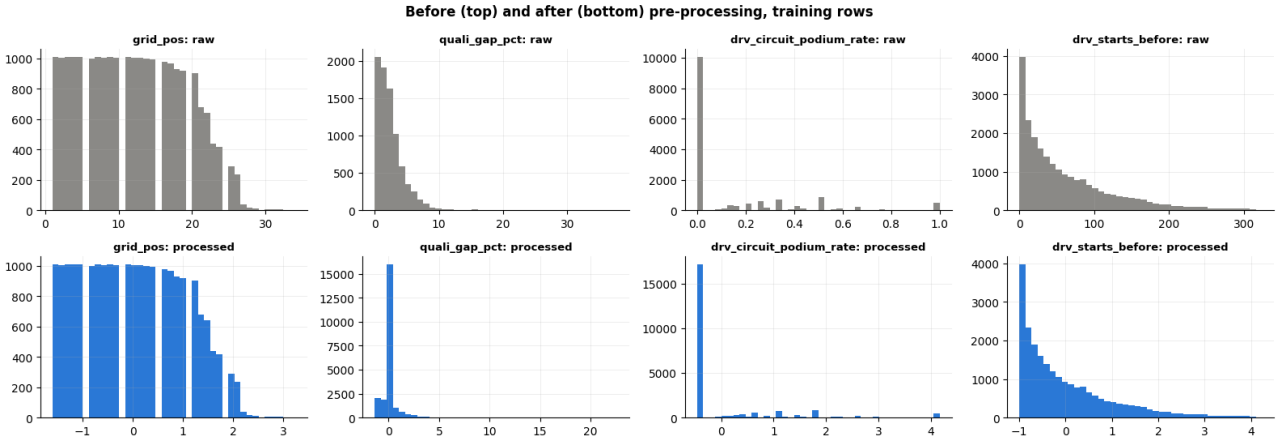


Figure 17. Effect of the pre-processing on the training data: distribution and range before (top) and after (bottom). Scaling changes the axis, not the shape; filled values appear as one bar at the median.

5.2 Preprocessing-order experiments

Each experiment changes **one** thing; same features, same logistic regression, scored on the validation seasons.

Pipeline	Val. PR-AUC	Val. ROC-AUC	What goes wrong
Correct: train-only statistics, flags before filling	0.7171	0.9275	–
A wrong: medians / scaler learned on all 1950–2024 rows	0.7175	0.9275	future seasons shape the inputs: the learned means move towards 2020–24 (more qualifying times, experience, reliability)
B wrong: gaps filled before the flags are added	0.7170	0.9256	all three flags become 0 for every row: 63.7% / 31.4% / 5.2% of rows lose the "was missing" information

Table 7. Preprocessing-order experiments.

Conclusion: the scores barely move (differences < 0.001, because 70 training seasons dominate the statistics and the validation seasons have almost no gaps), so the score alone cannot pick the order. We choose by correctness: A would make the test seasons no longer unseen, and B erases which values were real. The correct pipeline is used for every model.

6. Predictive modeling

6.1 Metrics and threshold

- **Primary metric: PR-AUC.** Podiums are rare (15%); what matters is ranking the real podium finishers at the top. PR-AUC measures exactly that and needs no threshold (random guessing ≈ 0.15). ROC-AUC looks high for every model (≈0.93), because separating the many obvious back-markers is easy.
- **Secondary metric: F1** at the chosen threshold, to judge the actual yes/no decisions (balancing false alarms and missed podiums). Precision, recall and ROC-AUC are reported as well.
- **Threshold:** for each model, the threshold with the highest F1 on the **validation** seasons, then used unchanged on test. All settings, epochs, thresholds and the best model are chosen on train / validation; **test is scored once**.

6.2 Baseline and modeling attempts

Every model was tried in several versions; each version is trained on the training seasons and compared on the validation seasons (PR-AUC), and the best one is kept.

Baseline

Logistic regression on `grid_pos` only, the simplest reasonable rule ("start further forward, more likely to podium"): training PR-AUC 0.499, validation **0.639**.

Logistic regression

A weight per input, summed and squeezed by the sigmoid into a probability. We tune `C`, how strongly the weights are kept small (a smaller `C` gives a simpler model that is harder to overfit).

C	Train PR-AUC	Val. PR-AUC
0.01	0.567	0.713
0.1	0.571	0.716
1	0.571	0.717
10 (chosen)	0.571	**0.717**

Table 8. Logistic-regression attempts: all four values score almost the same, so the model is not sensitive to `C`.

Training (0.57) is *below* validation (0.72) because the training eras are harder to predict (a third of starts ended in breakdowns, fields of up to 34 cars); that is not overfitting, which would show the opposite gap.

Random forest

300 trees, each trained on a random sample of rows and inputs, with their answers averaged. We tune the tree depth and the minimum number of drivers a final answer may be based on (leaf size): deeper trees and smaller leaves memorise more.

Max depth	Leaf 1: train / val.	Leaf 10: train / val.
4	0.587 / 0.720	0.587 / 0.721
6	0.629 / 0.720	**0.621 / 0.728** (chosen)
8	0.710 / 0.714	0.674 / 0.727
10	0.823 / 0.704	0.724 / 0.721
None	**1.000 / 0.672**	0.784 / 0.727

Table 9. Random-forest attempts (PR-AUC). With leaf 1, deeper trees push training towards 1.000 while validation falls to 0.672: overfitting. A leaf size of 10 prevents it.

Shallow neural network

18 inputs → ReLU hidden layer(s) → 1 sigmoid output; Adam, binary cross-entropy, batch 32; early stopping on validation PR-AUC (patience 15, best weights restored). Three versions, each changing one thing:

Run	Hidden layers	Early stopping	Epochs	Train PR-AUC	Val. PR-AUC
NN-A (chosen)	32	yes	34	0.596	**0.750**
NN-B	64 → 32	yes	36	0.625	0.743
NN-C	64 → 32	**no**	100	0.755	0.621

Table 10. Neural-network attempts: a bigger network does not help (A vs B); without early stopping the network overfits (B vs C).

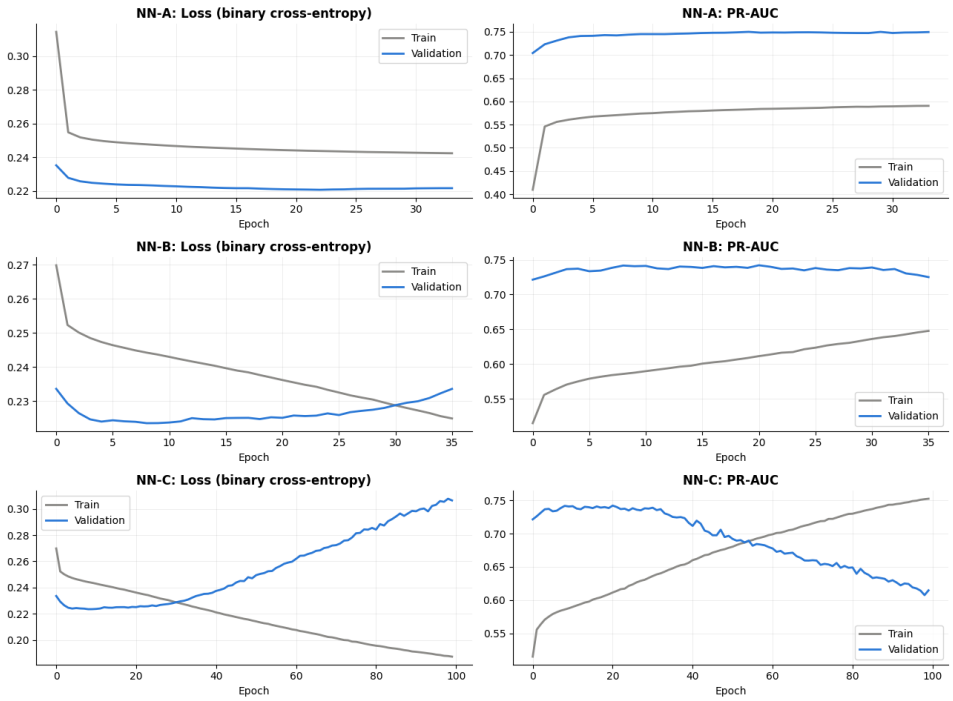


Figure 18. Training curves (loss and PR-AUC per epoch; grey = train, blue = validation). NN-C: training keeps improving while the validation loss turns upwards and validation PR-AUC falls, the overfitting early stopping prevents.

6.3 Final comparison on train, validation and the unseen test seasons

Model	Split	Thr.	ROC-AUC	PR-AUC	F1	Precision	Recall
Baseline (grid only)	train	0.433	0.884	0.499	0.544	0.546	0.543
	validation		0.909	0.639	0.692	0.692	0.692
	test		0.914	0.647	0.649	0.650	0.647
Logistic regression	train	0.248	0.900	0.571	0.586	0.490	0.729
	validation		0.927	0.717	0.705	0.645	0.778
	test		0.933	**0.715**	**0.672**	0.550	**0.863**
Random forest	train	0.320	0.910	0.621	0.606	0.551	0.674
	validation		0.929	0.728	0.699	0.659	0.744
	test		**0.939**	0.705	0.668	0.577	0.794
Neural network (NN-A)	train	0.470	0.908	0.596	0.550	0.606	0.504
	validation		0.930	**0.750**	**0.719**	**0.728**	0.709
	test		0.932	0.692	0.657	**0.652**	0.662

Table 11. All five metrics on train, validation and test. Thresholds were fixed on validation. Training scores are lower than validation for every model because the older training eras are harder to predict, not because of a problem with the split.

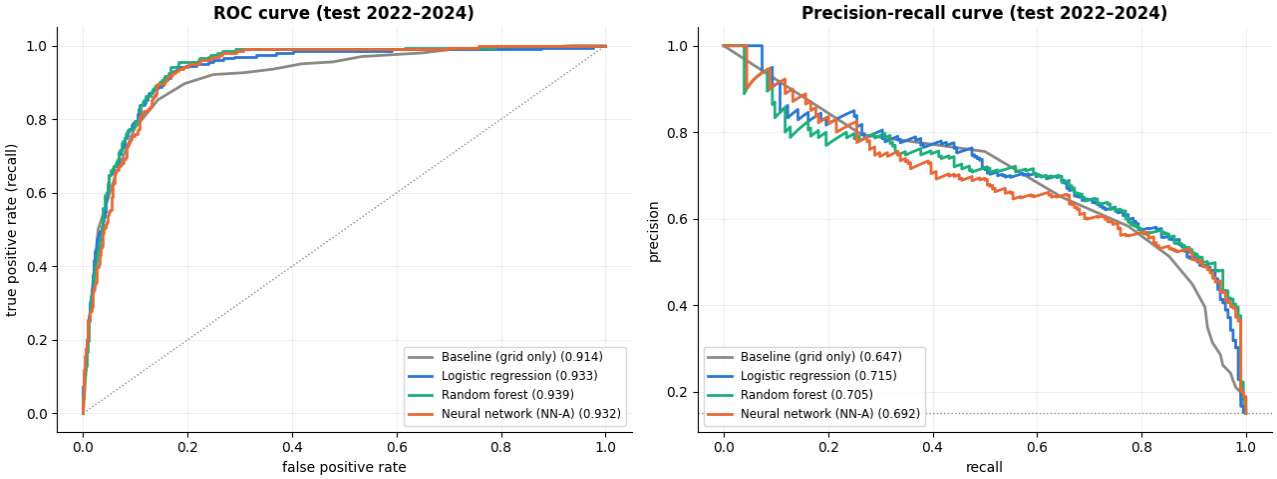


Figure 19. ROC and precision-recall curves on the test seasons 2022–2024 (dotted = random).

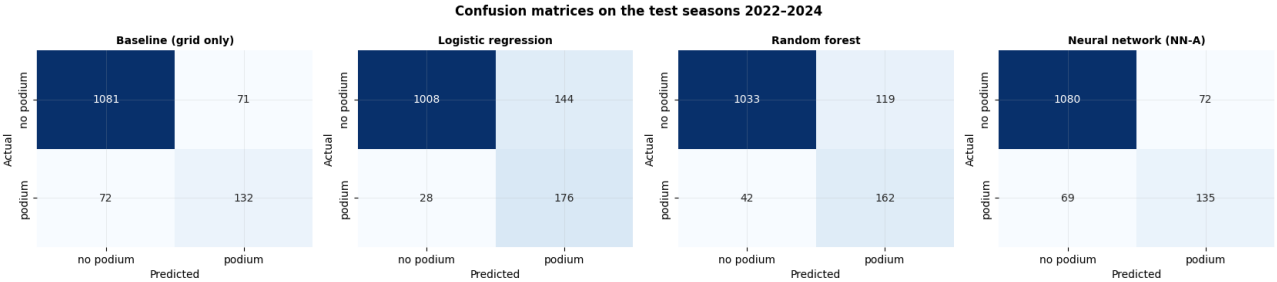


Figure 20. Confusion matrices on the test seasons (204 real podium finishers): the baseline catches 132 (71 false alarms), logistic regression 176 (144), the random forest 162 (119), the neural network 135 (72).

Reading the comparison

- **Every model beats the baseline** (test PR-AUC 0.647 → 0.692–0.715): form, standings and team features add real information beyond the starting position.
- **Validation picks the neural network** (0.750 vs 0.728 and 0.717), so by our rule it is the chosen model. **On the test seasons the order reverses:** logistic regression 0.715, random forest 0.705, neural network 0.692. The validation seasons contain only 117 podiums, and the network's lead did not carry over to the new rule era; its validation score is also slightly optimistic, because

early stopping used the validation seasons to pick its epoch. We do **not** switch models after seeing the test result (that would use the test set for a choice); we report it. The most stable model is logistic regression (0.717 validation, 0.715 test).

- **Thresholds set the trade-off:** logistic regression (0.25) catches the most podiums (recall 0.86) with the most false alarms; the network (0.47) is the most cautious (precision 0.65, recall 0.66).

7. Experiment: feature-group ablation

One group is removed at a time (its missing-value flag goes with it), the **chosen model (NN-A)** is retrained with exactly the same settings, and the PR-AUC drop is measured on validation (test shown as a check, not used for any choice).

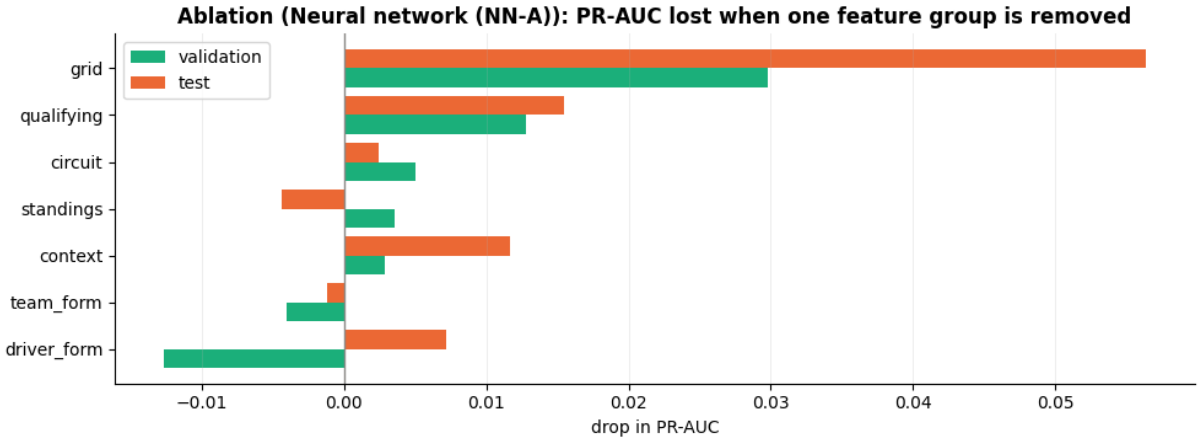


Figure 21. PR-AUC lost when one feature group is removed (aqua = validation, orange = test).

Group removed	Inputs	Val. PR-AUC	Drop (val.)	Test PR-AUC	Drop (test)
grid	1	0.720	**0.030**	0.635	**0.056**
qualifying	2	0.737	**0.013**	0.676	**0.015**
circuit	2	0.745	0.005	0.689	0.002
standings	7	0.747	0.004	0.696	−0.004
context	1	0.747	0.003	0.680	0.012
team form	3	0.754	−0.004	0.693	−0.001
driver form	2	0.763	−0.013	0.684	0.007

Table 12. Ablation (full model: validation 0.750, test 0.692).

- **Grid** is by far the most important group, and **qualifying** is second; both drop in the same direction on validation and test.
- The other groups show small changes in both directions: with 117 validation podiums and a network that changes slightly at every retraining, differences below ≈0.01 are noise.
- A small drop does **not** mean a group carries no information (driver form has correlation 0.46 with podium, Section 4.3): the other groups already contain it (a driver in form is usually also high in the standings and in a fast team, correlation ≈0.8). Ablation measures what a group adds **on top of the others**; Section 4.3 measured each feature on its own.

8. Explainability (XAI)

We explain the chosen neural network on the test seasons. Logistic regression is **intrinsically interpretable** (one weight per input), but the network is a **black box**, so we use **post-hoc, model-agnostic** methods that only use its inputs and outputs: **global** explanations (how the model works overall) and **local** ones (why one prediction was made). The explainers work on the inputs in real units (after filling, before scaling) through a function that scales and then predicts, which gives exactly the same probabilities.

8.1 Global: permutation importance

Each input is shuffled between the test drivers (10 repeats), keeping everything else, and the PR-AUC drop is measured.

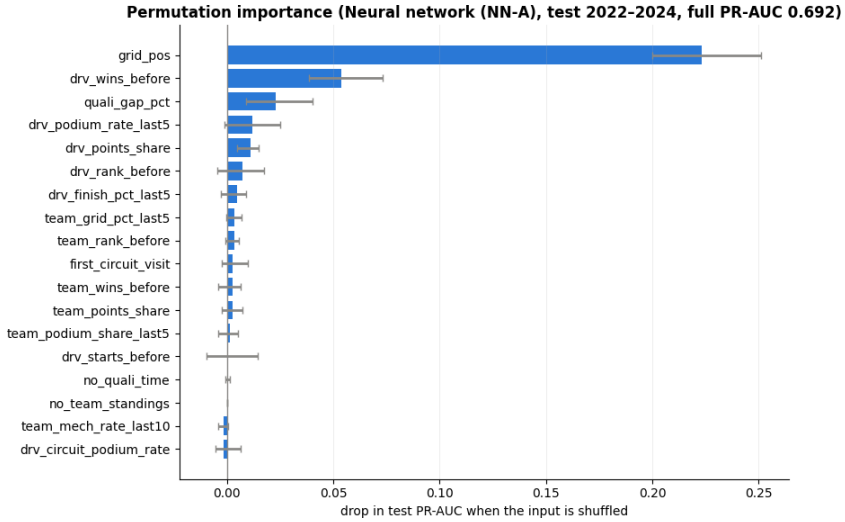


Figure 22. Permutation importance on the test seasons (bar = mean drop, line = smallest to largest of 10 shuffles).

- `grid_pos` **dominates** (drop ≈0.22), then `drv_wins_before` (0.05), `quali_gap_pct` (0.02) and `drv_podium_rate_last5` (0.01): the model uses **speed on the day** and **recent success**. The rest cost about 0.01 or less; the flags are ≈0 because they are almost always 0 in 2022–24.
- Unlike the ablation (which retrains), permutation keeps the trained model: *this* network relies on wins, while a network retrained without the standings can get the same information from form and grid.

8.2 Global: SHAP

SHAP divides one prediction fairly between the inputs ("sharing the prize"): prediction = average prediction of a reference group + one contribution per input. The reference is 100 driver-races from the **last ten training seasons (2010–2019)**; with all 1950–2019 rows, the flags would look important only because modern drivers *have* qualifying times.

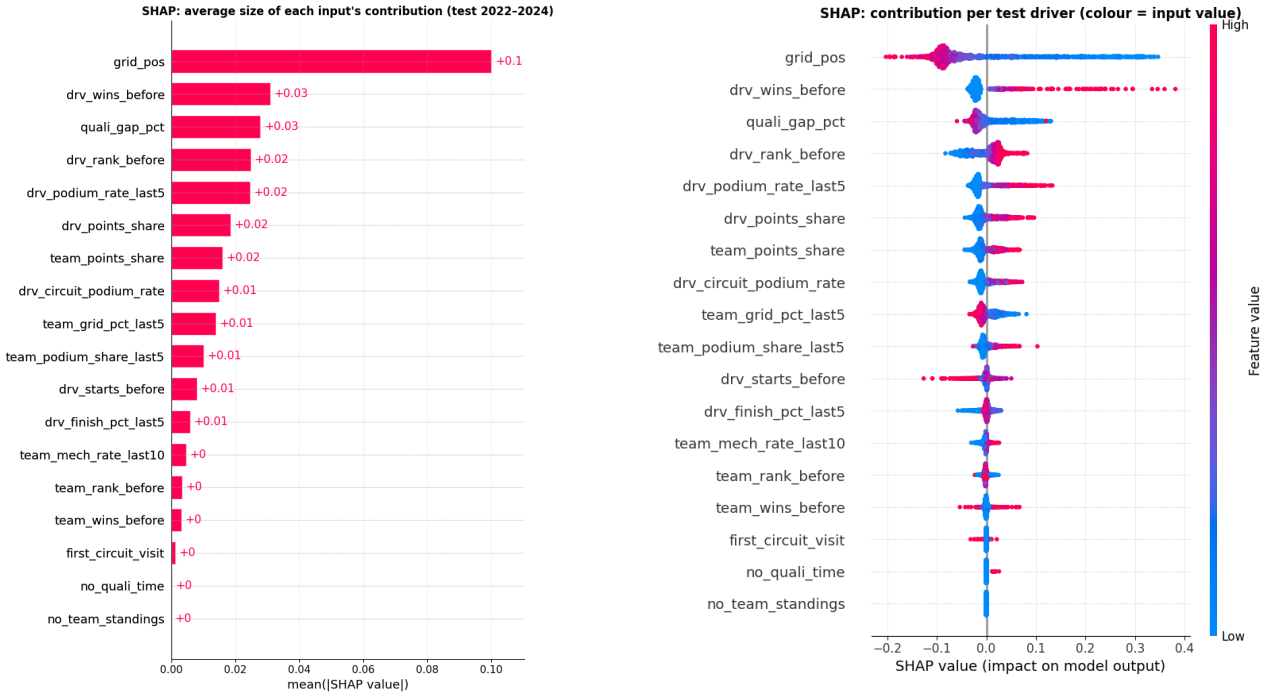


Figure 23. SHAP on the test seasons. Left: mean absolute contribution per input (reference average 0.182). Right: beeswarm, one dot per test driver; position = contribution, colour = input value (red high, blue low).

- **Same top three as permutation importance:** grid (≈0.10 on average), driver wins and qualifying gap (≈0.03), then driver standings and recent podium rate (≈0.02). Two methods agree, which makes the ranking more trustworthy.
- **Directions match F1 sense:** a front grid slot (blue) adds up to ≈+0.35; many wins and podiums (red) push up; a small qualifying gap and a fast car (blue) push up.
- A few overlapping inputs point the "wrong" way (e.g. a worse championship rank nudges the probability slightly up): rank, points share and wins carry the same information, so the model splits the credit and uses some as small corrections. They should not be read alone.

8.3 Local: one driver at a time (force, waterfall, LIME)

Two test drivers chosen by a **fixed rule** (not by the model's answers): the pole sitter of the first test race, and the podium finisher who started furthest back. For each: a SHAP force and waterfall plot, and **LIME**, which perturbs the driver's inputs, asks the network, and fits a simple surrogate model around this one driver.

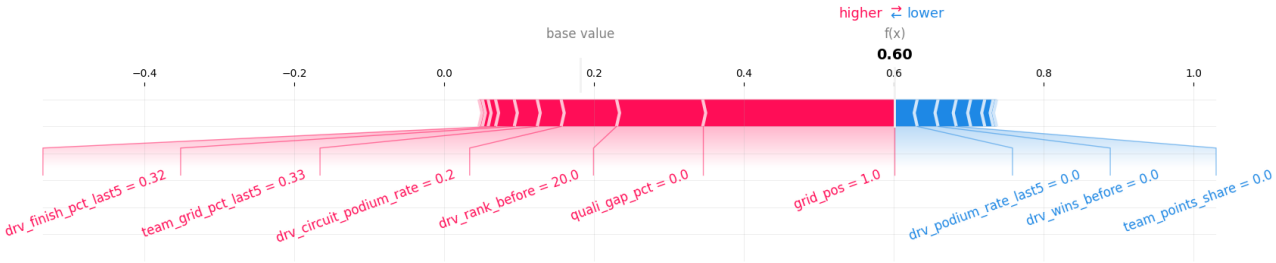


Figure 24. Case 1, Charles Leclerc, Bahrain 2022 (pole, won), SHAP force plot: red pushes up, blue pushes down; model probability 0.60 → podium.

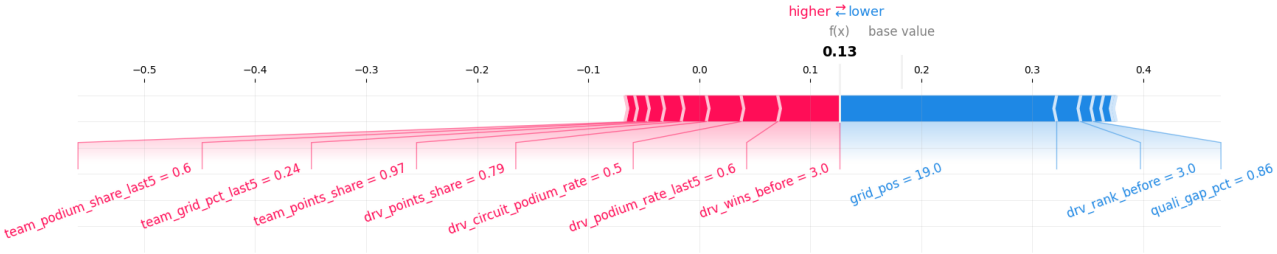


Figure 25. Case 2, Charles Leclerc, Abu Dhabi 2024 (qualified 14th, started 19th after a grid penalty, finished 3rd), SHAP force plot: probability 0.13 → no podium (a miss).

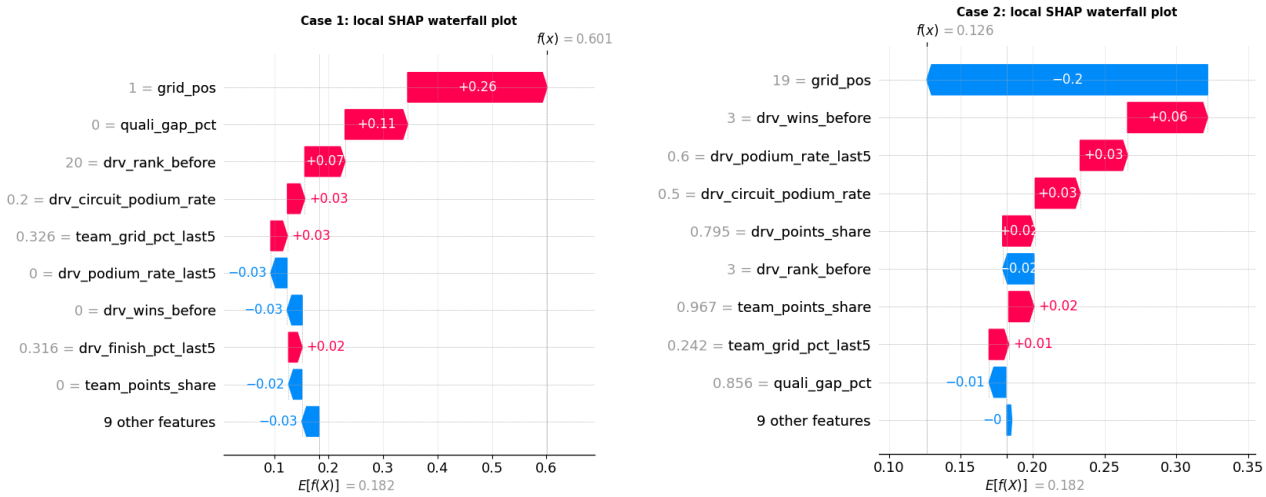


Figure 26. SHAP waterfall plots: Case 1 (left) and Case 2 (right), from the reference average (bottom) to the driver's probability (top).

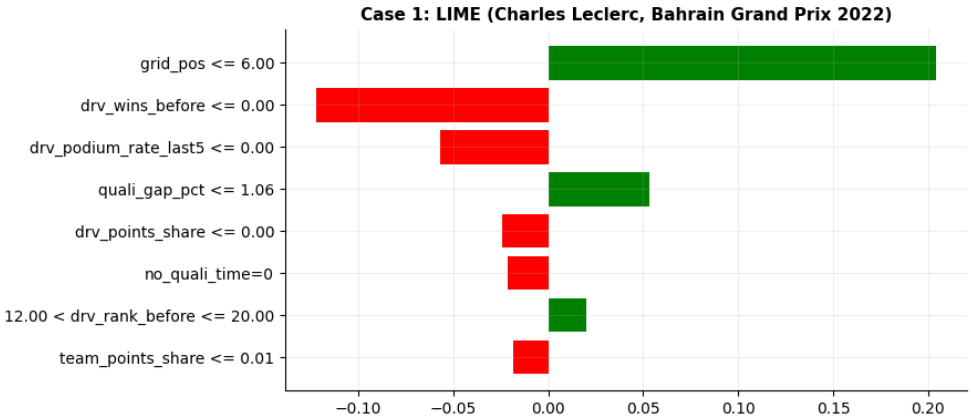


Figure 27. LIME explanation, Case 1 (Leclerc, Bahrain 2022): green conditions push towards podium, red away.

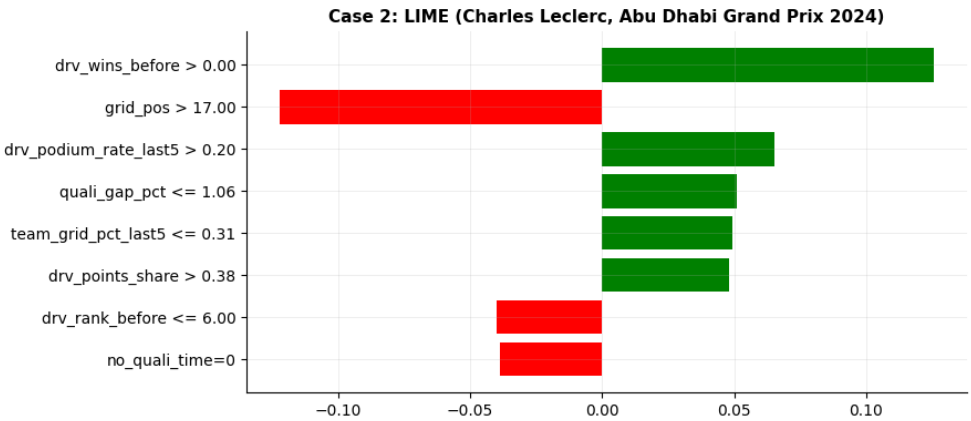


Figure 28. LIME explanation, Case 2 (Leclerc, Abu Dhabi 2024).

- **Case 1 (0.60, correct):** pole and the fastest lap are the largest pushes up; it is **round 1**, so all standings are zero and he had no podium in his last five races of 2021, which push down. The model trusts the grid and lap time over the weak form.
- **Case 2 (0.13, missed):** strong form pushes up (3 wins that season, podiums in 3 of his last 5 races, past podiums at this circuit, a fast car), but starting **19th** (≈−0.20) outweighs all of it together. The model does not know a fast car can overtake many cars; in its training data a P19 start almost never ends on the podium.
- SHAP and LIME agree on the main reasons; details differ because they answer slightly different questions (contribution vs a reference group, vs the effect of small changes around one driver). LIME's small minus for `no_quali_time=0` comes from its own perturbations.

8.4 Explanations are not causal effects

A feature being important to the model is **not proof that it causes podium finishes**; it only means the model relies on it to make predictions. Example: `drv_wins_before` is the second most important input, but wins do not cause the next podium: a **fast car and a good driver cause both** the earlier wins and the next podium. Likewise the grid slot and the podium both come largely from the car's speed; overlapping inputs share credit in ways that do not describe the real world; and giving a driver extra points would not make his car faster.

9. Inference function

`predict_podium(record)` predicts one new driver-race from its pre-race values: it rejects unknown input names; builds one row in the training column order (inputs left out or `None` become missing); applies **exactly the training pre-processing** (same flags, training medians, training scaler; nothing re-learned); asks the chosen model; compares with the validation-chosen threshold (0.47); and reports **which inputs were missing and what replaced them**.

Record	Missing inputs	Probability	Prediction	Actual
Complete: Leclerc, Bahrain 2022 (pole)	none	0.601	podium	1st
Partially missing: Bearman's debut, Saudi Arabia 2024 (P11)	driver podium rate, finish percentile, circuit history → training medians (0.0, 0.508, 0.0); `first_circuit_visit` set	0.046	no podium	7th

Table 13. Inference demonstrations.

The complete record returns the same probability as the evaluation pipeline (checked with an assert), so inference and evaluation are identical. The partial record still works and shows the user which values were filled, since a prediction built on filled values is less certain.

10. Known limitations

- **Historical qualifying-coverage drift:** qualifying times exist only from 1994 (patchy 1996–2002, complete from 2003), and team standings only from 1958. The flags let the models use those seasons, but for 64% of training rows the model learns without the qualifying signal.
- **Era sensitivity:** the models learn from 1950–2019 and are tested on a new rule era (2022–2024); all three full models score lower on test than on validation, and a future rule change could hurt them again. In practice they would be retrained on the newest seasons.
- **Small validation set:** 2020–2021 contain only 117 podiums, so validation scores are noisy; the network led on validation by 0.022–0.033 PR-AUC but was last of the three full models on test (its validation score is also slightly optimistic, because early stopping used the validation seasons to pick the epoch).
- **One driver at a time:** the model does not know a race has exactly three podiums; probabilities of one race do not add up to 3.
- **Information the data does not have:** weather, tyre strategy, safety cars, crashes and in-race failures are unknown before the start (Case 2 shows a recovery drive the model could not foresee).
- **Model-specific:** logistic regression cannot learn combinations unless they are built as features; the random forest overfits quickly with deeper trees; the network needs early stopping, and its exact numbers can differ slightly on another machine; a seed makes it repeatable on the same machine.
- **Data choices:** team renames start a new history (e.g. Toro Rosso → AlphaTauri → RB); nationality is only a proxy for "home"; ambiguous retirement labels are not counted as mechanical.